

Jamming Detection - LE910, ME910, ME310, NE310L2 series

Application Note

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1 Applicability Table

Table 1: Applicability Table

PRODUCTS	Platform Version ID	Technology
LE910 Cat1 SERIES	20	4G
LE910 V2 SERIES		
LE910Cx SERIES	25	
ME910C1 SERIES	30	
ML865C1 SERIES		
ME310G1 SERIES	37	
ME910G1 SERIES		
ML865G1 SERIES		
NE310L2 SERIES	43	

Note: Platform Version ID is a reference used in the document. It identifies the different SW versions, that is 20 for SW version 20.xx.xxx, 30 for SW version 30.xx.xxx, etc.

2 Introduction

2.1 Scope

This document covers methods used by Telit modules to reveal the jamming condition. It is divided in two main sections. The first one describes the methods used by Telit to reveal the jamming condition measuring all network carriers. The second one describes the methods to reveal the jamming condition using the serving cell information.

All the features and solutions described in this document are applicable to all variants listed in the applicability table

2.2 Audience

This document is intended for Telit customers, especially system integrators, interested to develop an application that needs notifications when the module is under jamming condition.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use: <https://www.telit.com/contact-us/>

For Product information and technical documents, visit: <https://www.telit.com>

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and Conditions

Refer to <https://www.telit.com/hardware-terms-conditions/>.

2.6 Disclaimer

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3 Jamming Detection on Platform Version ID 20, 25, 30, 37

This document describes the two methods implemented by Telit to recognize a jamming condition.

These two methods are both activated when the AT command **AT#JDRENH2** enables the jamming detection, refer to AT Commands Reference Guides [1], [2], [3], [4], or [5].

These two methods work independently each other except for the jamming detection state shared variable **JAMMING_STATE** that will be set to "JAMMED" if the module is under jamming condition and will be set back to "OPERATIVE" if the the module is no longer under jamming condition.

These two methods work in parallel: first method is more effective if the module is powered ON under jamming condition, in fact in this case the second method cannot detect the jamming condition; second method is more effective if the module goes under jamming condition when it is already registered to the network, in fact in this case the second method is faster than first method and first method is used as a backup if second method does not detect jamming condition.

When the module is in "OPERATIVE" state, the methods collect information of the serving cell carrier and the neighbor carriers. If one of the two methods detects that the module might be under jamming condition, it will check the device network registration and if the device is not registered too, then the **JAMMING_STATE** variable value will be changed to "JAMMED". The module shows this transition as is set through **AT#JDRENH2**.

When the module is in "JAMMED" state the methods check cyclically the network registration status and as soon as the device is registered again the **JAMMING_STATE** variable changes its value to "OPERATIVE". The module shows this transition as is set through **AT#JDRENH2**.

Note: **JAMMING_STATE**: is a discrete variable used by all jamming detect methods that can assume the state of "OPERATIVE" or "JAMMED".

Note: **OPERATIVE** is a condition that means that is possible, for the module, to communicate over the air.

Note: **JAMMED** is a condition that means that is not possible, for the module, to communicate over the air because there is a third device e.g. a "Jammer" that introduces interferences so powerfully and spread over the spectrum that do not allow other cellular devices to communicate.

3.1 First Method: Carriers Scan

The first method works on spectrum scan events both when the device is searching for a network and when the device is registered to the network. The method is based on the same algorithm for 2G, 3G and 4G networks.

3.1.1 Carrier Scan in 2G/3G/4G

Every time the module performs a spectrum scan, the Carrier Scan method checks (when JAMMING_STATE is "OPERATIVE") the power level of all carriers/frequencies and counts:

- in 2G how many carriers are with measured power above <sat2G> absolute threshold of RSSI (configurable using AT command **AT#JDRENH2**; only for devices supporting 2G)
- in 3G how many carriers are with measured power above <sat3G> absolute threshold of RSSI (configurable using AT command **AT#JDRENH2**; only for devices supporting 3G)
- in 4G how many carriers are with measured power above <P_RSSI_S4G> RSSI threshold level in dBm (configurable using AT command **AT#JDR4GCFG**).

The module might be under jamming condition if:

- in 2G the number of carriers that have a power level higher than <sat2G> is higher than <carrNum2G> (configurable using AT command **AT#JDRENH2**; only for devices supporting 2G)
- in 3G the number of carriers that have a power level higher than <sat3G> is higher than <carrN> (configurable using AT command **AT#JDRENH2**; only for devices supporting 3G)
- in 4G the number of carriers that have a power level higher than <P_RSSI_S4G> is higher than <Sampling_Number> (configurable using AT command **AT#JDR4GCFG**)

At this point, if the module registration status is NOT REGISTERED, the JAMMING_STATE variable will change its value to "JAMMED" and the module shows this transition as is set through **AT#JDRENH2**.

The Carrier Scan method will begin to work when jamming detection is activated by the **AT#JDRENH2** command. If jamming detection was previously activated, this method will be active since the module is switched ON and it will be ready to detect if the module is under jamming condition

Note: Carrier Scan method in 4G not applicable on Platform Version ID 20.

In 4G, Platform Version ID 25, 30 and 37, additional parameter `<Initial_Delay>` (configurable using AT command `AT#JDR4GCFG`) allows to filter out first invalid measures after power ON

3.2 Second Method: Serving Cell Monitoring

The second method collects a statistic of measured values of the serving cell when the device is registered to the network and works with different algorithms, one for 2G network, one for 3G network and one for 4G network.

3.2.1 2G Serving Cell Monitoring

When the module is in "OPERATIVE" state and is registered to a 2G cell, the algorithm collects the last *n* samples (e.g 8 samples - 10 samples) of `Gsm_RxLev` (Radio Signal Strength for the Serving Cell; 3GPP coded in 0-63) and calculates the average value `RxLev_av` (average of the last 10 `RxLev` measures; 3GPP coded in 0-63) for the samples collected that belong to the same cell as the last sample measured.

Then, the algorithm, calculates the `RxLev_Threshold` (3GPP coded in 0-63) a threshold value of `Gsm_RxLev` necessary to determine if the module is 2G jammed. The `RxLev_Threshold` is calculated with the formula showed below:

$$\text{RxLev_Threshold} = \text{RxLev_av} * (1 + \text{<pRxLevT2G>} / 100)$$

Where `<pRxLevT2G>` is a parameter configurable through AT command `AT#JDRENH2`.

In 2G network, if the module is jammed the `Gsm_RxLev` (3GPP coded in 0-63) will increase very quickly.

The algorithm decides that the module is jammed when the following two condition are both satisfied:

- `Gsm_RxLev > RxLev_Threshold`
- module registration status is NOT REGISTERED

Then the `JAMMING_STATE` variable will change its value to "JAMMED" and the module shows this transition as is set through `AT#JDRENH2`.

3.2.2 3G Serving Cell Monitoring

This clause applies only to devices supporting 3G.

When the module is in “OPERATIVE” state and it is registered to a 3G cell, the algorithm collects the last n samples (e.g 8 samples - 10 samples) of the two network parameters: EcN0 (Received energy per chip divided by the power density in the band) and RSCP (Radio Signal Code Power) and calculates the average value, respectively EcN0_av and Rscp_av for the samples collected that belong to the same cell as the last sample measured.

With the two average values calculated it is possible to calculate the thresholds EcN0_Threshold and Rscp_Threshold with the following formulas:

$$\text{EcN0_Threshold} = \text{EcN0_av} * (1 - \langle \text{pEcN0T3G} \rangle / 100)$$

Where $\langle \text{pEcN0T3G} \rangle$ is a parameter configurable through AT#JDRENH2 command.

$$\text{Rscp_Threshold} = \text{Rscp_av} * (1 - \langle \text{pRscpT3G} \rangle / 100)$$

Where $\langle \text{pRscpT3G} \rangle$ is a parameter configurable through AT#JDRENH2 command.

When the module is jammed in 3G the value of EcN0 will go quickly to zero and RSCP will remain near Rscp_av value or higher.

The algorithm decides that the module is jammed when the 3 points listed below happen for two consecutive samples:

- $\text{EcN0} < \text{EcN0_Threshold}$
- $\text{RSCP} > \text{Rscp_Threshold}$
- The module is not registered to any cell

If all these three conditions are satisfied for two consecutive EcN0 and RSCP samples then the JAMMING_STATE variable will change value to “JAMMED” and the module shows this transition as is set through AT#JDRENH2.

3.2.3 4G Serving Cell Monitoring

When the module is in “OPERATIVE” state and is registered to a 4G cell, the algorithm collects the last n samples (e.g 8 samples - 10 samples) of three network parameters: RSRP (in dBm), RSRQ (in dB) and RSSNR (in dB; if RSSNR is available) and calculates the average values, respectively: RSRP_av (in dBm), RSRQ_av (in dB), and RSSNR_av (in dB; if

RSSNR is available) for the samples collected that belong to the same cell as the last sample measured.

With the three average values calculated it is possible to calculate the thresholds RSRP_Threshold (in dBm), RSRQ_Threshold (in dB) and RSSNR_Threshold (in dB; if RSSNR is available) with the following three formulas:

$$\text{RSRP_Threshold} = \text{RSRP_av} * (1 - \langle \text{P_RSRP_T4G} \rangle / 100)$$

Where $\langle \text{P_RSRP_T4G} \rangle$ is a parameter configurable through AT command **AT#JDR4GCFG**.

$$\text{RSRQ_Threshold} = \text{RSRQ_av} * (1 + \langle \text{P_RSRQ_T4G} \rangle / 100)$$

Where $\langle \text{P_RSRQ_T4G} \rangle$ is a parameter configurable through AT command **AT#JDR4GCFG**.

$$\text{RSSNR_Threshold} = \text{RSSNR_av} * (1 - \langle \text{P_RSSNR_T4G} \rangle / 100)$$

Where $\langle \text{P_RSSNR_T4G} \rangle$ is a parameter configurable through AT command **AT#JDR4GCFG**.

When the module is jammed in 4G the RSRP (in dBm) value will at first increase shortly and then it will decrease very fast; RSRQ (in dB) value will decrease very fast and RSSNR (in dB) value will decrease very fast.

The algorithm decides that the module is jammed when the following conditions are satisfied:

- RSRP (in dBm) > RSRP_Threshold (in dBm)
- RSRQ (in dB) < RSRQ_Threshold (in dB) OR RSRQ (in dB) < -18 dB
- RSSNR (in dB) < RSSNR_Threshold (in dB) OR RSSNR (in dB) < 0 dB (if RSSNR is available)
- module registration status is NOT REGISTERED

Then the JAMMING_STATE variable will change its value to "JAMMED" and the module shows this transition as is set through **AT#JDRENH2**.

Note: On Platform Version ID 25, 30 and 37 $\langle \text{P_RSSNR_T4G} \rangle$ parameter is NOT present, hence RSSNR is NOT used in Jamming Detection algorithm.

3.3 Return to “OPERATIVE” State

When the module is in “JAMMED” state, the 2G or 3G or 4G serving cell monitoring algorithms collect no Gsm_RxLev, no EcN0, no RSCP, and no RSRP, RSRQ, RSSNR samples, but as soon as the module registration status becomes REGISTERED then the JAMMING_STATE variable changes its value to “OPERATIVE”. The module will show this transition as is set through **AT#JDRENH2**.

4 Jamming Detection on Platform Version ID 43

The jamming detection method described in this chapter is only available for NB IoT radio access technology. The jamming detection algorithm relies on band scan events which are triggered on foreground network search. Network search runs at module power on and in case of out of coverage.

The AT command **AT#JDRENH2** should be used to enable jamming detection and set the desired jamming reporting mode. **AT#JDR4GCFG** should be used to configure the jamming detection parameters. Both commands are documented in AT Commands Reference Guide [6].

4.1 Band Scan Method

A band scan implies the RSSI measurement across all EARFCN inside an LTE band. After one band scan is completed, the information about the Maximum and Minimum RSSI measurements are available (<PM_RSSI_MAX_S4G> and <PM_RSSI_MIN_S4G> respectively).

It is possible to calculate the Measured RSSI Average (<PM_RSSI_AVG_S4G>) and Measured RSSI Range (<RM_RSSI_S4G>) on each band scan.

$$\text{<PM_RSSI_AVG_S4G>} = (\text{<PM_RSSI_MAX_S4G>} + \text{<PM_RSSI_MIN_S4G>}) / 2$$

$$\text{<RM_RSSI_S4G>} = \text{<PM_RSSI_MAX_S4G>} - \text{<PM_RSSI_MIN_S4G>}$$

These two values can be used to recognize some peculiar network scenarios: the table below is meant to give a simplified overview of some of them as a function of RSSI Average and RSSI Range. Each value reported is purely indicative.

		RSSI Average	
		Low (< ~ -110 dBm)	High (> ~ -70 dBm)
RSSI Range	Low (< ~ 10)	Antenna detach	Jamming
	High (> ~ 30)	Poor coverage	Noise interference

The jamming detection algorithm compares the measured RSSI Average and Range values with the thresholds defined by AT#JDR4GCFG: <P_RSSI_S4G> and <R_RSSI_S4G>. The module might be under jamming if the conditions below are met for one single band scan result:

- <PM_RSSI_AVG_S4G> >= <P_RSSI_S4G>
- <RM_RSSI_S4G> <= <R_RSSI_S4G>

When jamming detection is active, the band scan order prioritizes the last LTE band used.

The jamming reporting method kicks in when the conditions are met, and it stops when the module is registered to the network.

To better evaluate the network scenario and tune the jamming detection thresholds for the best performance, AT#JDRNEH2=6 has been introduced. <mode> 6 returns the last 10 band scan results from the most recent to the oldest. Each line starts reporting the Jamming Detect Indicator, which is set to "1" if the jamming conditions for that LTE band are satisfied.

5 Acronyms and Abbreviations

Table 2: Acronyms and Abbreviations - General

Acronym	Definition
ARFCN	Absolute Radio Frequency Channel Number
AT	Attention command
BA	BCCH Allocation
BCCH	Broadcast Control Channel
CA	Cell Allocation
CBM	Cell Broadcast Message
CBS	Cell Broadcast Service
CCM	Current Call Meter
CLIR	Calling Line Identification Restriction
CTS	Clear To Send
CUG	Closed User Group
DCD	Data Carrier Detect
DCE	Data Communication Equipment
DCS	Digital Cellular System
DGPS	Differential GPS, the use of GPS measurements, which are differentially corrected
DNS	Domain Name System
DSR	Data Set Ready
DTE	Data Terminal Equipment
DTMF	Dual Tone Multi Frequency
DTR	Data Terminal Ready
GGA	GPS Fix data
GLL	Geographic Position – Latitude/Longitude
GLONASS	Global positioning system maintained by the Russian Space Forces
GMT	Greenwich Mean Time
GNSS	Any single or combined satellite navigation system (GPS, GLONASS and combined GPS/GLONASS)
GPRS	Global Packet Radio Service
GPS	Global Positioning System
GSA	GPS DOP and Active satellites
GSM	Global System Mobile
GSV	GPS satellites in view
HDLC	High Level Data Link Control
HDOP	Horizontal Dilution of Precision
IMEI	International Mobile Equipment Identity
IMSI	International Mobile Subscriber Identity
IP	Internet Protocol
IRA	International Reference Alphabet
IWF	Interworking Function
ME	Mobile Equipment
MO	Mobile Originated
MT	either Mobile Terminated or Mobile Terminal
NMEA	National Marine Electronics Association
NVM	Non-Volatile Memory
PCS	Personal Communication Service
PDP	Packet Data Protocol
PDU	Packet Data Unit

Acronym	Definition
PIN	Personal Identification Number
PPP	Point to Point Protocol
PUK	Pin Unblocking Code
RLP	Radio Link Protocol
RMC	Recommended minimum Specific data
RTS	Request To Send
SAP	SIM Access Profile
SCA	Service Center Address
SMS	Short Message Service
SMSC	Short Message Service Center
SMTP	Simple Mail Transport Protocol
TA	Terminal Adapter
TCP	Transmission Control Protocol
TE	Terminal Equipment
UDP	User Datagram Protocol
USSD	Unstructured Supplementary Service Data
UTC	Coordinated Universal Time
VDOP	Vertical dilution of precision
VTG	Course over ground and ground speed
WAAS	Wide Area Augmentation System

Table 3: Acronyms and Abbreviations - Specific

Acronym	Definition
EcN0	Received energy per chip divided by the power density in the band
Gsm_RxLev	Radio Signal Strength for the Serving Cell
RSCP	Received Signal Code Power
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
RSSI	Received Signal Strength Indicator
RSSNR	Signal-to-Noise Ratio
RxLev_Threshold	Threshold of RxLev in 2G Network
Sat2G	Starting Absolute Threshold 2G (RxLevel)
Sat3G	Starting Absolute Threshold 3G (RSSI)

6 Related Documents

For current documentation and downloads, refer to <https://dz.telit.com/>.

Table 4: Related Documents

S.no	Doc Code	Document Title
1	80446ST10707A	Telit LE910 V2 Series AT Commands Reference Guide
2	80529ST10815A	ME910C1/ML865C1 AT Commands Reference Guide
3	80617ST10991A	ME310G1/ME910G1/ML865G1 AT Commands Reference Guide
4	80586ST11192A	LE910Cx Linux AT Commands Reference Guide
5	80586ST11193A	LE910Cx ThreadX AT Commands Reference Guide
6	80668ST11067A	NE310L2 AT Commands Reference Guide

7 Document History

Table 5: Document History

Revision	Date	Changes
2	2025-01-13	New Document template Added NE310L2 to title and Applicability Table Added Chapter 4 Jamming Detection on Platform Version ID 43 Updated Related Documents section
1	2022-03-21	Updated to the new template; Legal Notices updated Platform Version IDs 25, 30 and 37 added General revision of various clauses
0	2015-11-10	Initial release

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